

Employee Attrition Prediction Using Machine Learning

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Project Title:

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1. Introduction

Employee attrition refers to employees leaving an organization due to resignation, retirement, or termination. High attrition can increase recruitment costs, reduce productivity, and negatively impact organizational performance. Therefore, predicting employee attrition helps organizations identify employees who are at risk of leaving and enables proactive retention strategies.

The objective of this project is to build machine learning models that predict employee attrition based on employee-related information.

2. Objective

The main objectives of this project are:

Perform data cleaning and preprocessing.

Explore the dataset using Exploratory Data Analysis (EDA).

Handle categorical variables using encoding techniques.

Handle class imbalance using SMOTE.

Train multiple machine learning model

Compare model performance using Accuracy, Precision, and Recall.

Identify the most important features influencing employee attrition.

3. Dataset Description

The project uses the IBM HR Employee Attrition dataset.

Initial Dataset

Number of Records: 1470

Target Variable: Attrition

Features include:

Age

BusinessTravel

Department

DistanceFromHome

Education

EducationField

Gender

JobRole

MaritalStatus

MonthlyIncome

OverTime

TotalWorkingYears

YearsAtCompany

YearsInCurrentRole

YearsWithCurrManager

and other HR-related attributes.

4. Data Preprocessing

The following preprocessing steps were performed.

4.1 Missing Values

The dataset was checked for missing values.

Observation

No missing values were found.

4.2 Duplicate Records

Duplicate records were checked.

Observation

No duplicate records were found.

4.3 Constant Features Removal

The following constant features contained only one unique value and were removed.

EmployeeCount

Over18

StandardHours

These features do not contribute to prediction.

4.4 Removing Identifier Feature

The EmployeeNumber feature was removed because it is only a unique identifier and has no predictive value.

4.5 Encoding

The target variable Attrition was encoded as:

No → 0

Yes → 1

All remaining categorical variables were encoded using Label Encoding.

4.6 Train-Test Split

The dataset was divided into:

Training Set (80%)

Testing Set (20%)

using stratified sampling.

4.7 Handling Class Imbalance

The training dataset was imbalanced.

SMOTE (Synthetic Minority Oversampling Technique) was applied only to the training data to balance the target classes.

4.8 Feature Scaling

StandardScaler was applied to normalize numerical feature values before model training.

5. Exploratory Data Analysis (EDA)

Several visualization techniques were used to understand the dataset.

5.1 Target Variable Distribution

The Attrition variable was visualized using a count plot.

Observation

The dataset is imbalanced.

Employees who stayed are significantly more than employees who left.

5.2 Distribution of Numerical Features

Histograms were plotted for important numerical variables.

Observations

Age is approximately normally distributed.

MonthlyIncome is positively skewed.

TotalWorkingYears is positively skewed.

YearsAtCompany is positively skewed.

5.3 Boxplot Analysis

Boxplots were created for numerical features.

Observations

Outliers were observed in:

MonthlyIncome

TotalWorkingYears

YearsAtCompany

These outliers were retained because they represent genuine employee information rather than data errors.

5.4 Correlation Analysis

A correlation heatmap was generated to study relationships between numerical variables.

Observation

Strong positive relationships were observed among:

JobLevel and MonthlyIncome

YearsAtCompany and YearsInCurrentRole

YearsAtCompany and YearsWithCurrManager

5.5 Categorical Feature Analysis

Bar charts were used to analyze the relationship between categorical variables and employee attrition.

The following features were analyzed:

Department

JobRole

MaritalStatus

Gender

OverTime

These visualizations provided insights into employee attrition across different categories.

6. Machine Learning Models

The following machine learning algorithms were trained.

Logistic Regression

Decision Tree

Random Forest

XGBoost

7. Model Evaluation

The models were evaluated using:

Accuracy

Precision

Recall

Model Comparison

Model	Accuracy	Precision	Recall
Logistic Regression	0.7993	0.4000	0.5106
Decision Tree	0.7517	0.3088	0.4468
Random Forest	0.7993	0.3421	0.2766
XGBoost	0.8265	0.4444	0.3404

8. Feature Importance

Feature importance analysis was performed using the Random Forest model.

The ten most influential features were identified based on importance scores.

9. Results Discussion

The performance comparison indicates that:

XGBoost achieved the **highest Accuracy** (82.65%), making it the best-performing model overall.

XGBoost also obtained the highest Precision (44.44%), meaning it was the most accurate when predicting employees who would leave.

Logistic Regression achieved the **highest Recall** (51.06%), indicating it detected a larger proportion of employees who actually left the organization.

Decision Tree showed the lowest overall performance among the evaluated models.

Random Forest achieved good accuracy but had a relatively low Recall, meaning it missed many employees who actually left.

These results show that while **XGBoost performed best overall**, **Logistic Regression was more effective** at identifying employees who were likely to leave.

10. Conclusion

This project successfully developed machine learning models to predict employee attrition using employee demographic, financial, and work-related information.

The dataset was cleaned, preprocessed, and explored using various EDA techniques. Categorical variables were encoded, class imbalance was handled using SMOTE, and feature scaling was applied before model training.

Four machine learning models were trained and evaluated.

Among all models:

XGBoost achieved the highest Accuracy (82.65%) and Precision (44.44%).

Logistic Regression achieved the highest Recall (51.06%).

Therefore, XGBoost is recommended as the final model because it provided the best overall predictive performance for this dataset.

11. Future Work

Future improvements may include:

Hyperparameter tuning using GridSearchCV or RandomizedSearchCV.

Trying additional ensemble algorithms such as LightGBM or CatBoost.

Performing feature selection to improve model performance.

Applying cross-validation for more reliable evaluation.

Deploying the final model as a web application using Flask or Django.

Overall Outcome

The project demonstrates how machine learning techniques can be used to predict employee attrition and assist organizations in making data-driven human resource decisions. By identifying employees who are at higher risk of leaving, organizations can implement targeted retention strategies, improve employee satisfaction, and reduce turnover costs.